SMART CAFETERIA MANAGEMENT SYSTEM

INTEGRATION TESTING REPORT

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Date: 2026-03-04T06:02:15.718Z
Duration: 15.16s Server:
http://localhost:5000

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SECTION 1: SYSTEM COMPONENTS IDENTIFIED

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Backend Stack:
- Runtime: Node.js + Express.js
- Database: MongoDB (Mongoose ODM) via Atlas
- Authentication: JWT (jsonwebtoken) + bcryptjs
- Validation: express-validator
- CORS: cors middleware enabled

Frontend Stack:
- Framework: React Native (Expo)
- HTTPClient: Axios
- StateMgmt: React Context (AuthContext)
- Config: EXPO_PUBLIC_API_URL → /api

Database Models:
- User
- Booking
- Slot
- SlotTemplate
- MenuItem
- Menu
- CrowdData
- CrowdPrediction
- AlertLog
- DemandForecast

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SECTION 2: INTEGRATION POINTS

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Auth → User Model → JWT Middleware
Frontend → Axios → Express Routes
Routes → Controllers → MongoDB
Role Middleware → Protected APIs

SECTION 3: TEST EXECUTION RESULTS

Summary:
Total Tests: 70

Passed:70
Failed:0
PassRate:100%

SECTION 4: PASSED SCENARIOS (70)

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[PASS] [1. System Initialization] Server health check (GET /)
[PASS] [1. System Initialization] API base path accessible (GET /api/auth)
[PASS] [1. System Initialization] CORS headers present
[PASS] [1. System Initialization] Menu slots endpoint accessible (public)
[PASS] [1. System Initialization] Menu items endpoint accessible (public)
[PASS] [2. Authentication Flow] Register new student
[PASS] [2. Authentication Flow] Prevent duplicate email registration
[PASS] [2. Authentication Flow] Login with valid student credentials
[PASS] [2. Authentication Flow] Reject login with wrong password
[PASS] [2. Authentication Flow] Reject login with non-existent email
[PASS] [2. Authentication Flow] Access protected route GET /api/auth/me with valid token
[PASS] [2. Authentication Flow] Update user profile
[PASS] [2. Authentication Flow] Login as admin
[PASS] [3. Admin Workflow] Admin registers a new staff member
[PASS] [3. Admin Workflow] Staff member logs in
[PASS] [3. Admin Workflow] Admin fetches all staff members
[PASS] [3. Admin Workflow] Admin adds a new menu item
[PASS] [3. Admin Workflow] Admin updates menu item
[PASS] [3. Admin Workflow] Admin fetches analytics
[PASS] [3. Admin Workflow] Admin fetches slot-wise analytics
[PASS] [3. Admin Workflow] Admin fetches staff performance
[PASS] [3. Admin Workflow] Admin fetches waste tracking
[PASS] [3. Admin Workflow] Admin fetches sustainability report
[PASS] [3. Admin Workflow] Admin triggers data backup
[PASS] [4. Student Workflow] Student fetches menu items
[PASS] [4. Student Workflow] Student fetches available slots
[PASS] [4. Student Workflow] Student fetches menu for specific slot
[PASS] [4. Student Workflow] Student places a booking
[PASS] [4. Student Workflow] Student fetches active tokens
[PASS] [4. Student Workflow] Student fetches booking details by ID
[PASS] [4. Student Workflow] Student modifies booking items
[PASS] [4. Student Workflow] Student fetches all booking history
[PASS] [5. Staff Workflow] Staff fetches queue for slot
[PASS] [5. Staff Workflow] Staff marks booking as serving
[PASS] [5. Staff Workflow] Staff marks booking as served
[PASS] [5. Staff Workflow] Staff calls next token in queue
[PASS] [6. Booking Cancellation] Create booking for cancellation test
[PASS] [6. Booking Cancellation] Student cancels booking successfully
[PASS] [6. Booking Cancellation] Cannot cancel already cancelled booking
[PASS] [7. Middleware Integration] JWT middleware blocks access without token
[PASS] [7. Middleware Integration] JWT middleware blocks invalid token
[PASS] [7. Middleware Integration] JWT middleware blocks malformed token
[PASS] [7. Middleware Integration] Student cannot access admin routes
[PASS] [7. Middleware Integration] Student cannot access staff routes
[PASS] [7. Middleware Integration] Staff cannot access admin routes
[PASS] [7. Middleware Integration] Staff cannot create bookings (student-only route)
[PASS] [7. Middleware Integration] Admin cannot create bookings (student-only route)
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[PASS] [7. Middleware Integration] Student cannot add menu items (admin-only)
[PASS] [7. Middleware Integration] Staff cannot register other staff (admin-only)
[PASS] [8. Error Handling] Registration fails with missing fields
[PASS] [8. Error Handling] Booking fetch with invalid ID returns error
[PASS] [8. Error Handling] Booking fetch with non-existent ID returns 404
[PASS] [8. Error Handling] Update non-existent menu item returns 404
[PASS] [8. Error Handling] Delete non-existent menu item returns 404
[PASS] [8. Error Handling] Delete non-existent staff returns 404
[PASS] [8. Error Handling] Menu for non-existent slot returns 404
[PASS] [8. Error Handling] Booking with non-existent slot fails
[PASS] [8. Error Handling] Staff mark-serving non-existent booking returns 404
[PASS] [8. Error Handling] Staff mark-served non-existent booking returns 404
[PASS] [9. Database CRUD Validation] User: Read user via /api/auth/me
[PASS] [9. Database CRUD Validation] Slot: Read all slots via /api/menu/slots
[PASS] [9. Database CRUD Validation] MenuItem: Read all items via /api/menu/items
[PASS] [9. Database CRUD Validation] MenuItem: Delete test item via DELETE /api/menu/items/:id
[PASS] [9. Database CRUD Validation] MenuItem: Verify deleted item no longer in list
[PASS] [9. Database CRUD Validation] Booking: Verify booking history contains entries
[PASS] [10. Cross-Module Integration] Booking → Slot: Booking updates slot currentBookings
[PASS] [10. Cross-Module Integration] Auth → Booking: Token from auth works for booking routes
[PASS] [10. Cross-Module Integration] Admin → Staff → Queue: Complete staff lifecycle
[PASS] [10. Cross-Module Integration] Menu → Slot → Booking: Full ordering workflow data consistency
[PASS] [11. Cleanup] Admin deletes test staff member
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SECTION 5: FAILED SCENARIOS (0)

None — All tests passed!

SECTION 6: SECURITY ISSUES & OBSERVATIONS

[CRITICAL] JWT_SECRET missing initially
[MEDIUM] No admin user deletion API
[LOW] Weak password policy
[LOW] No refresh token mechanism
[INFO] Registration endpoint has no rate limiting

END OF INTEGRATION TESTING REPORT